

Miraj Chamara Samarakkody

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SUMMARY

Assistant Professor of Mathematics at Tougaloo College with a Ph.D. from Texas Tech University. Current research focuses on manifold learning and formal proof verification, building on a background in pure and applied differential geometry and variational problems. Principal Investigator on NIH- and NSF-funded research projects, with published work in differential geometry and open-source contributions to mathematical software.

RESEARCH INTERESTS

Manifold learning and geometric methods in data science; formal proof verification and formalization of mathematics in Lean 4; variational problems, geometric analysis, and mathematical physics; mathematical software development in Julia.

EDUCATION

Ph.D. in Mathematics, Fall 2021 – Spring 2025
Texas Tech University, Lubbock, TX
Advisors: Dr. Hung Tran (Chair), Dr. Álvaro Pámpano (Co-chair)

B.Sc. (Honors) Special in Mathematics, 2014 – 2018
University of Peradeniya, Sri Lanka
Advisor: Dr. Shelton Perera

PROFESSIONAL EXPERIENCE

Assistant Professor, Spring 2025 – Present
Department of Mathematics and Computer Science, Tougaloo College, MS

Graduate Part-Time Instructor, Fall 2024
Department of Mathematics and Statistics, Texas Tech University, TX

Research Assistant, Spring 2022 – Summer 2024
Department of Mathematics and Statistics, Texas Tech University, TX
Mentors: Dr. Hung Tran and Dr. Álvaro Pámpano

Graduate Teaching Assistant, Fall 2021
Department of Mathematics and Statistics, Texas Tech University, TX

Lecturer, 2019 – Fall 2021
Department of Mathematics, University of Moratuwa, Sri Lanka

SHORT VISITS

Research Mentor, Virtual Math Circle, Department of Mathematics, Louisiana State University
Summer 2025

TECHNICAL SKILLS

Programming Languages: Python, Julia, C

Mathematical Software: Mathematica, MATLAB

Typesetting: LaTeX, Typst

Theorem Proving: Lean 4

Design and Media: Blender, Adobe Illustrator, Adobe Premiere Pro

Productivity: Microsoft Office

TEACHING

Tougaloo College

- MAT103 - College Algebra; Summer 2025, Summer 2026
- MAT221 - Calculus I; Fall 2025, Spring 2026, Fall 2026
- MAT222 - Calculus II; Spring 2025, Spring 2026, Fall 2026
- MAT316 - Differential Equations; Fall 2025; Fall 2026
- MAT326 - Introduction to Probability; Fall 2025
- MAT414 - Modern Algebra; Spring 2025
- MAT426 - Advanced Calculus; Spring 2025, Fall 2025, Fall 2026
- MAT429 - Complex Variables; Spring 2026, Fall 2026
- MAT434 - Theory of Mathematical Statistics; Spring 2025, Spring 2026

Texas Tech University

- MATH 1452 - Calculus II with Applications; Fall 2024

GRANTS

- **Principal Investigator**, Institutional Development Award (IDeA), National Institute of General Medical Sciences, National Institutes of Health, Grant Number P20GM103476. Amount: \$29,317. Project: Multivariate Statistical Analysis and Machine Learning Approaches for Heart Disease Prediction and Risk Factor Identification.
- **Recipient**, HBCU UP Implementation Project, Award Number 2510537.
- **Principal Investigator**, SCALE Program (Partnership with Purdue University). Amount: \$73,094.77. Project: Microelectronic Workforce Development Project for Radiation-Hardening.

FELLOWSHIPS

- **Dr. Shelby Hildebrand Graduate Math Fellowship**, Texas Tech University, 2023 – 2024. Amount: \$10,000.
- **AT&T Graduate Fellowship**, Texas Tech University, 2021 – 2025. Amount: \$16,000.

AWARDS AND HONORS

- **Create Possible Scholarship**, Texas Tech University, 2022.
- **University Prize for Academic Excellence**, University of Peradeniya, 2018.
- **Peradeniya University Alumni Australia Victoria Chapter Scholarship for Academic Excellence in Mathematics**, 2015.

PUBLICATIONS

5. M. Samarakkody, “Physics-Informed Neural Networks for Computing the Morse Index of the Critical Catenoid,” Submitted.
4. M. Samarakkody. “Formalizing the Classical Isoperimetric Inequality in the Two-Dimensional Case.” Submitted.
3. Y. Li, E. Güler, M. Toda, M. Samarakkody. “Geometric Properties of Hypersurfaces of Right Conoids in Minkowski Spacetime.” Submitted.
2. A. Pámpano, M. Samarakkody, H. Tran. “Closed p -Elastic Curves in Spheres of $\mathbb{L}^3$.” Journal of Mathematical Analysis and Applications, 542(2):129147, 2025.
1. S.R.S.M.C. Samarakkody, P.G.R.S. Ranasinghe, A.A.S. Perera. “Geometric Behavior of a Certain Class of Quotients of Finite Blaschke Products.” International Conference on Multidisciplinary Engineering (ICMME), Post Graduate Institute of Science, University of Peradeniya, Vol. 1, 2019.

SOFTWARE

3. **IsoperimetricInequality** - Lean 4 formalization of the classical isoperimetric inequality in the two-dimensional case.
Repository: github.com/mirajcs/IsoperimetricInequality
DOI: 10.5281/zenodo.20258500
2. **ParametricCurves.jl** - Julia package for analyzing parametric curves.
Repository: github.com/mirajcs/ParametricCurves.jl
DOI: 10.5281/zenodo.20288830
1. **SymKit.jl** - Julia package for symbolic computation, supporting algebraic manipulation, differentiation, simplification, evaluation, and limits with Unicode pretty-printing.
Repository: github.com/mirajcs/SymKit.jl
DOI: 10.5281/zenodo.20294029

PRESENTATIONS

Conference Talks

3. “Closed p -Elastic Curves in 2-Space Forms.” AMS Sectional Meeting (Meeting No. 1198), University of Texas at San Antonio, September 14–15, 2024.
2. “Closed p -Elastic Curves in Spheres of Lorentz-Minkowski Space.” Texas Analysis and Mathematical Physics Symposium, Texas A&M University, February 10, 2024.
1. “Geometric Behavior of a Certain Class of Quotients of Finite Blaschke Products.” ICMME, Post Graduate Institute of Science, University of Peradeniya, March 2019.

Posters

2. “Formalizing the Isoperimetric Inequality in Lean 4.” XX Red Raider Minisymposium - Geometric Analysis and Applications, Texas Tech University, April 25, 2026.
1. “Closed p -Elastic Curves in 2-Space Forms.” Texas Geometry and Topology Conference (68th Meeting), Texas A&M University, College Station, November 8–10, 2024.

Seminars

4. “From Manifold to Machines: Formalizing Differential Geometry in Lean.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, April 24, 2026.
3. “On Some Variational Problems in Curves and Surfaces.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, January 21, 2025.
2. “On Some Variational Problems in Differential Geometry.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, November 1, 2023.
1. “Minimal Surfaces.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, November 2, 2022.

LEADERSHIP AND COMMUNITY INVOLVEMENT

- Member, Commission on Animal Care - Institutional Review Board (IRB), Tougaloo College, 2025 – 2026.
- Judge, Kincheloe Research Symposium, Tougaloo College.
- Counselor, Summer Chess Camp 2024, Texas Tech University.
- Judge, Arts and Humanities Conference 2023, Texas Tech University.
- Mentor, Math Circle 2023 – 2024, Department of Mathematics and Statistics, Texas Tech University.

PROFESSIONAL MEMBERSHIPS

- Member, American Mathematical Society, 2021 – 2025.
- Member, SIAM Texas Tech University Chapter, 2021 – 2025.